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Magnetic Geophysical Modeling of Ophiolitic Bodies in Northwest Saudi Arabia – Implications for Natural Hydrogen Exploration

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Abstract

Saudi Arabia, located on the Arabian tectonic plate, is a promising area for exploring sites for natural hydrogen production and accumulation due to the broad existence of ophiolitic ultramafic rocks, especially serpentinites. These rocks, recognized as accreted forearc fragments, are exposed in the western and northwestern segments of the Precambrian Arabian Shield. The present study aims to model the geometry and depth of two ophiolitic suites in NW Saudi Arabia by analyzing airborne magnetic data. The airborne magnetic data were analyzed using edge-detection algorithms (magnetic field derivatives) to study the distribution of lineaments corresponding to rock contacts and tectonic structures, and a lineament density map was produced. The magnetic susceptibility of rock samples collected during field surveys in the broader study area was measured in the lab. The measured values were used as constraints for the 3D inversion of the magnetic data. A 3D geophysical model was constructed and used to determine the tectonic regime of the subsurface in relation to the distribution of the ophiolitic bodies. The lineament map presented the distribution of the tectonic structures, while the lineament density map indicated highly fractured areas. These areas are suitable for the infiltration of rainwater, which creates aquifers into the ophiolitic bodies. These highly fractured areas could serve as water carriers for serpentinization processes at suitable temperatures and provide conduits for vertical migration to possible traps. Serpentinization, a hydrothermal reaction between water and ultramafic rocks, typically occurs at temperatures greater than 200°C because of the oxidation state of reduced iron in the minerals. Considering an average regional temperature gradient, the suitable depth for serpentinization in the area is estimated to be more than 7 km. The 3D models made

by constraining their boundaries with respect to the surrounding rock formations show the distribution and geometries of the ophiolitic masses continuing to the required depths, exhibiting fractures suitable for hydrogen generation and subsequent upward migration. However, hydrogen accumulations within highly fractured areas within the ophiolitic bodies cannot be ruled out. This study provides insights into the potential for natural hydrogen production and migration from ophiolitic massifs and could be an analog for areas where thick sediments cover the Arabian Shield, providing traps for potential hydrogen accumulation. The results highlight the importance of integrating geologic knowledge with geophysical insights to model the distribution of these masses and can aid future hydrogen exploration projects in Saudi Arabia.

Introduction

In recent decades, numerous natural hydrogen (H2) seeps have been documented across various locations globally (Zgonnik, 2020; Moretti et al., 2021). The main geotectonic contexts for these seepages include serpentinized ultramafic rocks at mid-ocean ridges, compressional zones with ophiolitic nappes, and Precambrian crystalline shields. H2 gas seeps, associated with an ophiolitic origin, have been identified in Mali (Maiga et al., 2024; Hutchinson et al., 2024), Oman (Vacquand et al., 2018), and Albania (Truche et al., 2024). In Saudi Arabia, large ophiolitic ultramafic rocks are found in the northern Precambrian Arabian Shield (Johnson and Kattan, 2012), where the potential for H2 production through serpentinized rocks has been assessed by previous studies (Saidy et al., 2024).

Extracting H2 from ultramafic rocks presents several challenges that must be overcome for its commercial viability. More specifically, during the exploration phase, the sporadic and often diffuse nature of hydrogen occurrences makes it difficult to identify economically viable deposits, while the lack of established, effective exploration techniques tailored to this unique resource makes the detection of the resources even harder. The generation of hydrogen itself is highly sensitive to geological conditions; while serpentinization can occur broadly, optimal hydrogen production rates are observed within a narrow temperature window of 200°C to 315°C, with lower or higher temperatures significantly hindering the process (e.g., McCollom and Bach, 2009; Zgonnik, 2020). Furthermore, the specific mineralogical composition of the ultramafic rock, particularly its iron content, and the availability and chemistry of water are critical factors influencing the amount and rate of hydrogen generated. Once formed, hydrogen's small molecular size and high mobility make it prone to leakage from subsurface reservoirs, underscoring the critical need for effective geological traps and seals to ensure accumulation. Understanding the complex migration pathways of hydrogen through the subsurface and how it accumulates in these traps remains an active area of research (Loiseau et al., 2025). Additionally, the presence of subsurface microbial communities that consume hydrogen poses a biological challenge, potentially depleting reserves before extraction (Zgonnik, 2020). Even when hydrogen is successfully located, it often coexists with other gases like carbon dioxide, necessitating complex and costly separation processes to achieve the purity required for various applications. Finally, the long-term behavior of natural hydrogen reserves and the efficiency of extraction technologies on a large, commercial scale are still largely unproven, with no widespread commercial production currently established (Rosa et al., 2025; Han et al., 2025). Establishing the appropriate infrastructure for hydrogen transport also presents a significant economic and logistical challenge. Overcoming these geological, technical, and economic obstacles through continued research and development is crucial for unlocking the full potential of H2 as a sustainable energy source.

Geophysical methods offer an effective means for subsurface characterization. The integration of various geophysical data provides better insights into geological formations compared to traditional methods like geological mapping, borehole drilling, or using a single geophysical method. This approach reduces interpretive uncertainty and enhances the economic viability of exploration efforts. While H2 explorations in ophiolites have primarily relied on geochemical analysis of core samples (Ramirez et al., 2023; Lefeuvre et al., 2024), geophysics plays a crucial role in assessing potential H2 resources. Geophysical data can be utilized to construct conceptual subsurface models that can guide detailed surveys and inform exploratory drilling. The magnetic method, particularly effective for modeling ophiolitic bodies (Michels et al., 2020; Li et al., 2023), detects variations in magnetic field intensity linked to the different magnetic susceptibility of rocks (Blakely, 1995). The method is especially useful for the identification of serpentinized formations, which often present high magnetic susceptibility (Oufi et al., 2002; Maffione et al., 2014). While the magnetic method allows for the quick and cost- effective survey of extensive areas (Fairhead, 2016), it has limitations, including lower resolution compared to seismic methods and susceptibility to external noise from anthropogenic structures. Also, the non-uniqueness of the produced results requires the incorporation of supplementary information, like geological data or the application of additional geophysical techniques, to enhance the reliability of the constructed models. To minimize these challenges, the application of specialized data processing, filtering, and modeling techniques is essential for accurate interpretation.

This study used 3D inversion of airborne magnetic data to map the subsurface shape and depth of ophiolitic structures at Jabal Wask and Jabal Ess areas in northwest Saudi Arabia (Fig.1). To make the models more accurate, we measured the magnetic susceptibility of surface rock samples and used the measurements to constrain the inversion. The findings were validated with existing geological information from previous studies. The analysis also focused on identifying linear features such as rock contacts and tectonic elements, particularly those related to ophiolites, which are important for H2 accumulation. These linear features likely correspond to weak zones that affect groundwater circulation and the tectonic environment of the area. Enhanced fluid flow along these fault zones or at lithological contacts creates conditions that may facilitate serpentinization (a reaction where ultramafic rocks interact with water), resulting in the release of hydrogen gas. This makes these areas promising for potential H2 production.

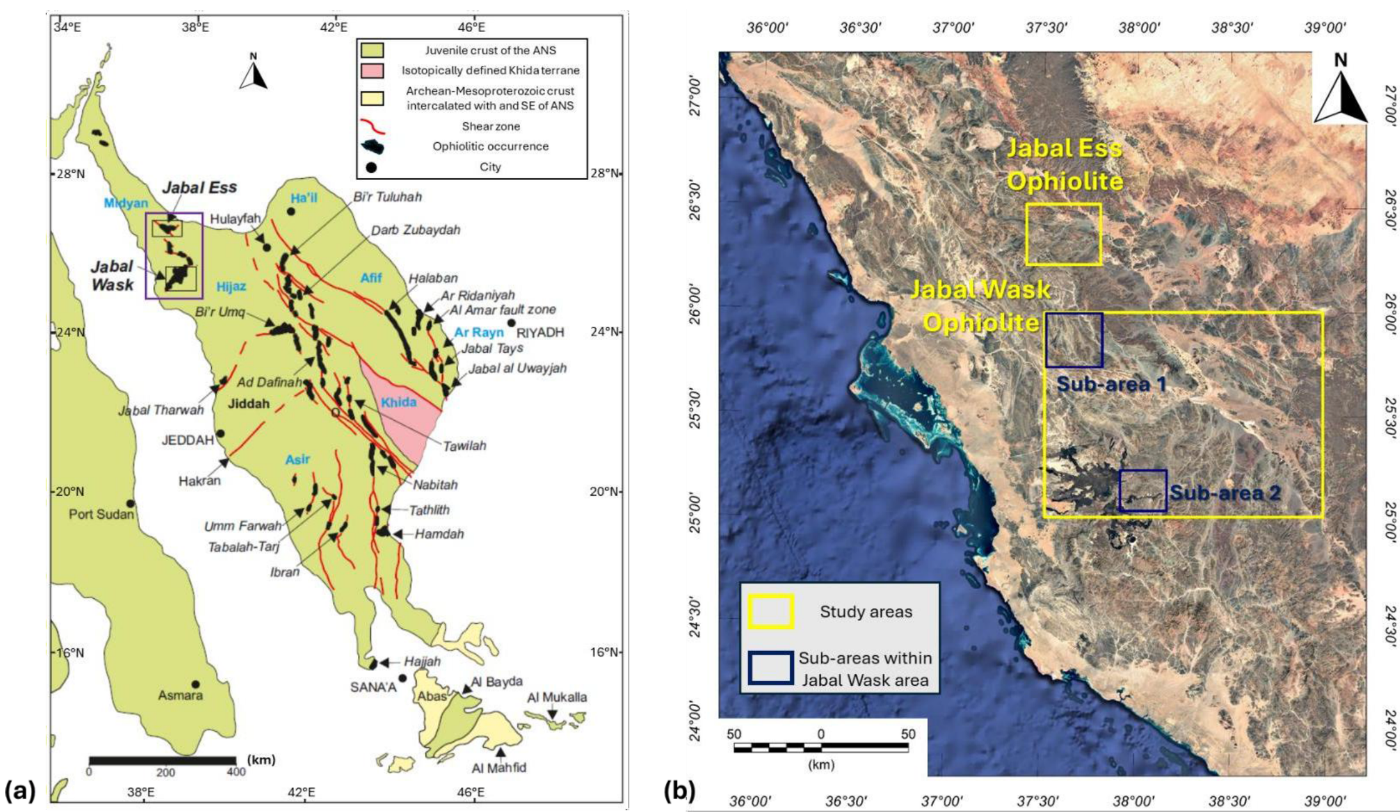

Figure 1—(a) Map of the wider ANS region (modified from Johnson and Kattan, 2012), highlighting the locations of ophiolites and the study areas of Jabal Ess and Jabal Wask in the northwestern part; (b) Satellite image zoomed in on the study area (Jabal Ess and Jabal Wask – yellow rectangles, sub-areas 1 and 2 within Jabal Wask – purple rectangles) (after Chavanidis et al., 2025).

Geology of the study area

The Arabian-Nubian Shield (ANS) lies in the northern segment of the East African Orogen (Fig. 1a) and represents the largest preserved segment of a 900-550 Ma crust on Earth (Patchett and Chase, 2002). ANS largely consists of stacked thin-skinned nappes resulting from oblique convergence and the amalgamation of intra-oceanic arcs (Stern, 1994; Johnson et al., 2011). Alkaline plutons were emplaced within the ANS between 640 and 550 Ma (Johnson et al., 2011; Robinson et al., 2014).

The Arabian Shield in Saudi Arabia is made up of several tectonostratigraphic terranes, separated by zones containing ophiolites (Stoeser and Camp, 1985). One such terrane is the Midyan terrane, located in the northwestern part of the country (Fig. 1a). The Midyan terrane, along with the Eastern Desert of Egypt and Sinai, forms the Midyan-Eastern Desert-Sinai (MEDS) composite terrane. This larger terrane is bordered to the south by the Yanbu suture zone (Kozdrój et al., 2018). This suture zone is significant because it represents where an ancient ocean basin closed, and the MEDS terrane collided with the Hijaz terrane (Whattam et al., 2024).

The ophiolites in this wide region are remnants of an 890-675 Ma oceanic lithosphere (Johnson and Kattan, 2012), indicative of island arcs suturing and terrane merging (Stoeser and Camp, 1985; Pallister et al., 1988; Johnson et al., 2004). The Jabal Ess ophiolite is a complex with an age of 761 ± 27 to 782 ± 36 Ma and 706 ± 11 to 780 ± 11 Ma, based on geochronological analyses of its host and intruding rocks (Claesson et al., 1984; Pallister et al., 1988; Whattam et al., 2024). The ophiolite, situated on the Yanbu suture, is bordered North and South by shear zones that converge in the East. To the West, it intersects with the northwest-trending Da'ban and Durr fault systems. Mafic to ultramafic formations extend into the Sahluj mélange and the Jabal Wask ophiolite (Ledru and Augé, 1984; Johnson et al., 2004). The latter exhibits significant serpentinization, similar to Jabal Ess, suggesting a common origin of the two complexes.

Data and Methodology

The magnetic data comes from a comprehensive dataset of airborne surveys that includes the Arabian Shield and the Phanerozoic cover in Saudi Arabia. These surveys were originally conducted between 1965 and 1966, and the flights were carried out at 300 m above ground with a distance between lines of 800 m (Zahran et al., 2003). Following initial corrections like leveling and IGRF correction, the data were processed and gridded at a 200 m cell size, preserving higher resolution and eliminating artifacts at the survey boundaries. For this study, the Saudi Geological Survey (SGS) provided the Total Magnetic Intensity (TMI) and Reduction to the Pole (RTP) maps. The RTP data for Jabal Wask and Jabal Ess are shown in Fig. 2. For analyzing the magnetic data, we utilized the Oasis montaj software. To validate the airborne data analysis results, samples were collected from the ophiolites in the study area, specifically from locations with unique mineralogical and structural features. The magnetic susceptibility of these samples was measured both in the field with a portable meter and confirmed later with laboratory measurements under controlled conditions, allowing for correlation between magnetic responses and specific rock types.

Magnetic data analysis can reveal linear features that indicate variations in geotectonic structures or filling materials, such as faults and dikes, due to the different magnetic properties that they exhibit (Fairhead, 2016). These lineaments are essential for understanding lateral geological discontinuities related to the tectonic regime and fluid dynamics, affecting groundwater movement and hydrogeological systems (e.g., Bolós et al., 2022). Such structures can create conditions favorable for H2 deposit formation and trapping. More specifically, to identify linear features, we used the CET (Centre for Exploration Targeting) Grid Analysis tool, which effectively maps lineament distribution using advanced algorithms (Holden et al., 2008; Tsepav, 2020), and manual corrections after data enhancements. Magnetic data interpretation is significantly assisted using techniques that emphasize distinct geophysical characteristics. One effective method is calculating magnetic field derivatives, which accurately delineate the edges of the anomalous sources and reveal features that might otherwise be missed (Fairhead, 2016). The tilt derivative filter, for instance, is a powerful tool for highlighting anomalies, regardless of whether they originate from shallow or deep sources. In our analysis, the tilt derivative was calculated from the RTP data to better identify geological lineaments (linear features, often indicative of faults or fractures). To ensure the accuracy of the results, these were cross-referenced with CET grid analysis, with manual corrections made to remove lineaments not aligned with zero- value contours, ensuring only relevant features were retained in the final map. Finally, to analyze how lineaments are spatially distributed and to understand their relationship to ophiolites, a lineament density map was created. This was achieved using the Line Density tool of QGIS software.

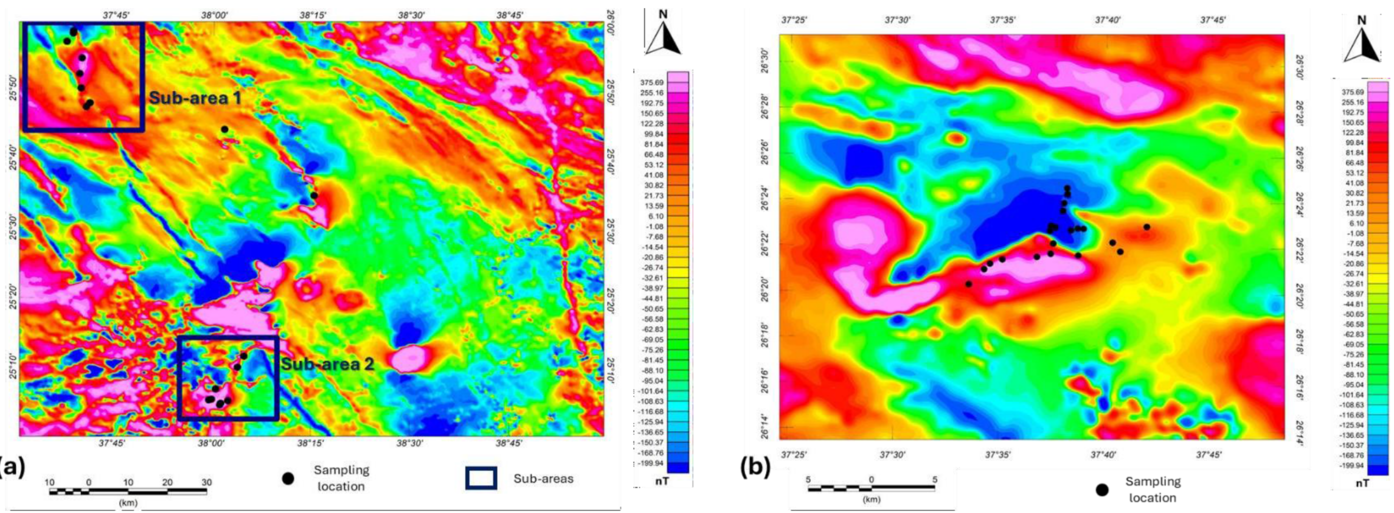

Figure 2—The RTP magnetic field in (a) Jabal Wask and (b) Jabal Ess study areas. The boxes in (a) indicate the two sub-areas within Jabal Wask that this study focuses on, while the black dots correspond to locations where samples from the ophiolitic outcrops were collected during the field survey (after Chavanidis et al., 2025).

For the 3D magnetic susceptibility inversion of RTP data, the VOXI Earth Modeling tool was employed. VOXI makes use of the Cartesian cut cell method. This technique divides the model area into prismatic elements, allowing for effective handling of irregular topography, thus leading to reduced computational complexity and faster processing time (Ingram et al., 2003). The inversion algorithm begins with a preliminary model that typically does not conform to the measured geophysical data and refines it through iterations to achieve a simpler model that approximates the geological reality. The study areas were fractioned into meshes of 33 × 33 × 10 cells, 50 × 44 × 15 cells, and 40 × 33 × 11 cells (for the two areas of Jabal Wask and Jabal Ess, at resolutions of 1000 m and 500 m, correspondingly), reaching depths of about 5.1 km for sub-area 1 of Jabal Wask and Jabal Ess and 4.6 km for sub-area 2 of Jabal Wask. Constraints on magnetic susceptibility values were set between 0.001 SI and 0.1 SI for both areas, using the measured magnetic susceptibility of the collected ophiolitic samples.

Results

The lineament density map is presented in Fig. 3. Overall, it shows mid- to high lineament density regions. Fewer lineaments are present in the Jabal Ess area compared to the two sub-areas in Jabal Wask, with a denser distribution in its central part.

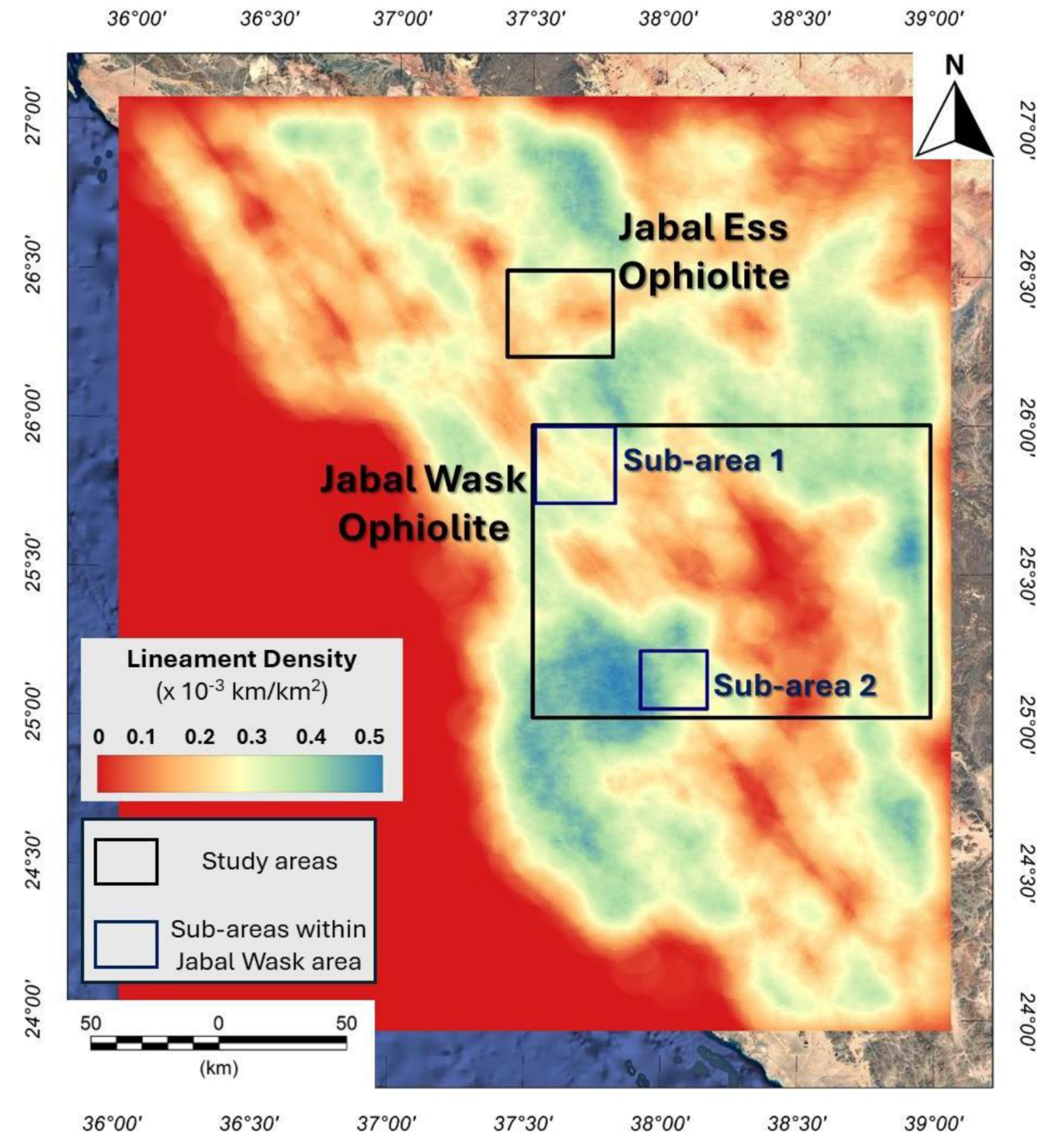

Figure 3—Lineament density map for the wider Jabal Wask and Jabal Ess areas (after Chavanidis et al., 2025).

The 3D modeling results for Jabal Ess are presented in Fig. 4. The pink-colored bodies indicate formations with higher magnetic susceptibility in comparison with their surrounding environment, ranging between 0.002-0.1 SI.

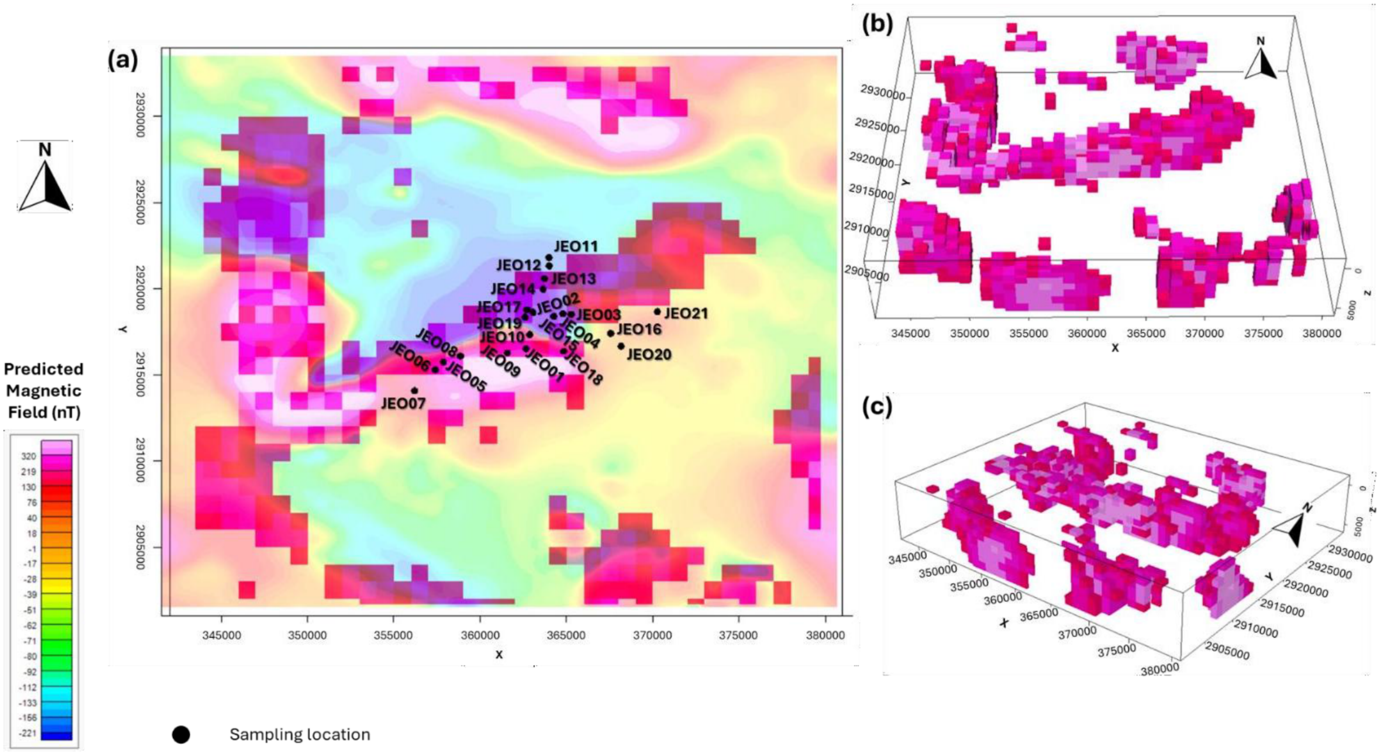

Figure 4—The 3D model of the Jabal Ess area, displaying bodies that mainly represent ophiolitic formations. (a) Top view, overlaid by the calculated magnetic field and the sites of the field samples (dots); (b) and (c) Lateral views (after Chavanidis et al., 2025).

Discussion

The magnetic field maps indicate that high anomalies are well aligned with the distribution of ophiolites, which typically produce strong magnetic signals. These anomalies may also suggest the presence of ophiolitic bodies in the subsurface, beneath less magnetic formations. The study of the lineament density map indicates that the ophiolitic structures in the Jabal Wask area likely enhance subsurface water circulation and H2 accumulation in the western part; however, more intact rock formations are expected in the central and eastern parts. For the Jabal Ess area, higher lineament density is encountered within the ophiolitic masses, yet no clear link between lineament density and ophiolitic distribution is evident. While a dense fracture network facilitates groundwater movement in these formations, factors like tortuosity, porosity, and permeability also play crucial roles in groundwater flow and must be considered.

The 3D modeling indicated that the distribution of magnetized rocks aligns closely with the ophiolitic masses in the Jabal Ess and Jabal Wask areas, and it can be used to estimate the depth and lateral extent of the latter. Typically, surface ophiolitic outcrops present lower magnetic susceptibility than their deeper sections; for this reason, we utilized higher susceptibility values for the modeling compared to those measured in the samples. This difference is likely due to the compaction of rocks and the presence of highly magnetic minerals at greater depths, which are often altered and eroded at the surface. For estimating hydrogen production from these masses, a detailed modeling approach is necessary. This model should address specific temperature and pressure gradients, the serpentinization degree, and the water-to-rock ratio; factors that are crucial for hydrogen generation through serpentinization (McCollom and Bach, 2009; Klein et al., 2013).

Ophiolites are naturally fractured ultramafic rocks usually formed by ocean floor extension and subsequent relocation. Fractures enhance water-rock interactions (Zgonnik, 2020; Milkov, 2022). During serpentinization, they undergo hydration and alteration, frequently due to groundwater in nearby aquifers, allowing minerals to form along fracture surfaces. The optimal temperature for serpentinization is between

200°C and 315°C (McCollom and Bach, 2009; Klein et al., 2013; Hutchinson et al., 2024); 200°C is estimated to occur at 7 km depth based on local temperature gradients in our study area. However, laboratory tests at 55 and 100°C produced notable H2 amounts from ophiolitic rocks (Mayhew et al., 2013). Interconnected fault and fracture networks manage fluid flow, with faults acting as key conduits for deeper fluids, thus enhancing serpentinization (Saidy et al., 2024). Surface lineaments and fracture zones are often associated with H2 gas seepages, as in the case of Oman, while characteristic geological features (e.g., Fairy Circles) are also attributed to H2 seeps at the surface (Zgonnik, 2020; Moretti et al., 2021). Such features have not been identified in the Arabian Shield, though. Alkaline springs are key indicators of serpentinization, where ophiolitic rocks alter and produce hyperalkaline fluids with high pH levels. In our study area, no springs have been reported, with the closest located 150 km to the East (Bazuhair et al., 1990). These springs exhibit pH levels between 8.35 and 8.45, with sodium comprising 50-90% of the major cations, while sulfate and chloride dominate the anions, and are encountered in an alkaline basalt area that was formed about 10 Ma due to Red Sea spreading. The water composition in our study area is also expected to be affected by the underlying basalts (Johnson and Kattan, 2012).

The ophiolitic rocks in the Jabal Ess and Jabal Wask areas outcrop at the surface, mainly as serpentinized peridotites. Serpentinization is stronger in sheared zones, where fault planes and related fractures facilitate the movement of fluids, allowing them to interact with the surrounding rock. Geophysical modeling indicated the presence of substantial, deep-seated structures beneath these two ophiolites, which suggests potential for H2 generation, migration, and storage. The ophiolites are thought to expose iron-rich minerals to water at depths of around 7 km, potentially producing significant amounts of H2. Furthermore, fractures and faults may serve as pathways for H2 to rise vertically, while a dense network of fractures within the host rock could act as potential reservoirs or traps for H2 accumulation; however, petrophysical properties, such as porosity and permeability, should also be considered.

Conclusions

The geological and tectonic characteristics of the Jabal Ess and Jabal Wask ophiolites in northwestern Saudi Arabia play a crucial role in shaping subsurface permeability, fluid migration routes, and the presence of hydrogen. A comprehensive understanding of these processes requires the integration of advanced subsurface modeling, geochemical analysis, and geophysical surveys. The way forward after this study would involve a progression from broad-scale identification to more focused and direct exploration, and eventually, if successful, to resource assessment and potential development. To achieve these, future research focusing on drilling targeted boreholes for the assessment of the petrophysical properties and magnetic susceptibility will enhance subsurface modeling. Detailed geochemical analyses of borehole samples and groundwater are essential for understanding natural hydrogen production and accumulation, wh ile geophysical logging will provide insights into rock characteristics that influence water movement. Utilizing high-resolution 3D geophysical surveys can improve structural reconstructions. The combination of magnetic, gravity, magnetotelluric, and reflection seismic methods will allow for the comprehensive modeling of promising structures and identification of natural hydrogen reservoirs, whereas geoelectrical and electromagnetic surveys can further characterize fracture networks and study the hydraulic properties. In general, the integration of geophysical, geological, geochemical, and economic perspectives will enhance energy and resource exploration in the Arabian Shield's promising ophiolitic complexes.

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